

Edge-AI and Multimodal Sensor Fusion–Based Smart Stick for Visually Impaired Individuals with Real-Time Obstacle Detection

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Abstract— Independent mobility for visually impaired individuals is often limited by the inability of conventional assistive tools to provide early and reliable awareness of surrounding obstacles. While recent smart navigation aids incorporate sensing and artificial intelligence, many existing solutions depend on cloud computation or single-modality sensing, resulting in latency, privacy concerns, and reduced robustness in real-world environments. This work proposes a fully on-device smart stick based on Edge Artificial Intelligence and multimodal sensor fusion for real-time obstacle detection. A lightweight TinyML-based convolutional neural network processes visual input on a XIAO ESP32S3 microcontroller, while ultrasonic and infrared sensors supply complementary distance and proximity information. These modalities are combined using a decision-level fusion strategy to enhance detection reliability under varying environmental conditions. Experimental validation conducted in indoor and semi-outdoor settings demonstrates an average detection accuracy of 0.89, with precision and recall values of 0.88 and 0.86, respectively. The system achieves inference latency below 125 ms and operates with low power consumption suitable for prolonged use. The results confirm the feasibility of deploying multimodal Edge-AI perception on resource-constrained assistive devices.

Keywords: Edge Artificial Intelligence, Assistive Navigation, Multimodal Sensor Fusion, TinyML, Embedded Vision Systems, Real-Time Obstacle Detection

I. INTRODUCTION

Visual impairment has a significant influence on the self-sufficiency and self-safety, particularly in the context of a new or a dynamically changing environment. According to global health reports, millions of people rely on assistive devices to move around in the space of daily activities, and in these cases, unidentified barriers, including furniture, stairs, or moving objects, may cause an accident and undermine confidence. Traditional mobility aids, such as white canes,

provide tactile feedback over a short distance only and in doing so require physical contact with the obstructions. Despite being essential tools, their sensing abilities are limited, which explains why it is important to have smart assistive technologies that can sense the surroundings that cannot be accessed instantly and provide early warnings.

The last ten years have seen the emergence of tremendous advancements in artificial intelligence and embedded systems, which has triggered the growth of smart mobility support, incorporating electronic sensors and computational intelligence. The systems that involve vision, especially, would offer high contextual details in terms of identification of object types and space definition in the surrounding. Similarly, the distance sensors, including ultrasonic and infrared sensors are commonly used due to their affordability and durability. However, many of the current assistive solutions are based on a single sense, making them vulnerable to noise in the environment like a low-light environment, reflective surfaces, or sensor noise. This means that in real life conditions, obstacle detection reliability is not coherent.

A number of studies have explored the use of camera-based obstacle detection using deep learning models and have obtained promising results in controlled laboratory settings. In spite of these achievements, the methods often rely on cloud computing or supercomputers, therefore, adding to the latency, higher energy usage, and privacy issues. Other systems use ultrasonic or infrared sensors only, and they provide quick responses but do not provide semantic understanding of the environment. Such paradigms normally yield binary warnings without differentiating between different obstacles disparate types and contextual risks, hence restricting their use in challenging maneuvering exercises. Based on this, there is still a disparity between the high-accuracy perception systems and the real-life assistive

devices that can be used in the daily life. The other drawback of existing smart stick systems is that they do not give sufficient attention to embedded deployment requirements. Most of the proposed designs fail to consider the issues related to implementing machine learning models on microcontrollers with low power, which deter scalability and sustainability. In addition, some of the studies because of lack of effective sensor-fusion strategies have consequences of redundant sensing or higher rates of false-positive. This inadequacy amplifies the need to have a single system which facilitates accurate perception, efficiency in computations and energy consumption without sacrificing real-time responsiveness.

Feasibility studies play a critical role in the development of assistive technologies, particularly when deploying intelligent perception systems on resource-constrained embedded platforms. Prior feasibility-oriented research has demonstrated that while advanced sensing and AI techniques show promise in controlled environments, practical constraints such as power consumption, latency, hardware limitations, and user safety significantly influence real-world deployment [6], [9], [10]. These studies emphasize the need to validate not only detection accuracy but also real-time performance and energy efficiency under realistic operating conditions. Motivated by these findings, the present work focuses on evaluating the feasibility of integrating Edge-AI vision and multimodal sensor fusion within a low-power smart stick, ensuring that the proposed system can operate reliably in everyday assistive navigation scenarios.

The system uses an ultrasonic and infrared sensing, a TinyML-optimized convolutional neural network, and is run on a resource-constrained XIAO ESP32S3 microcontroller. Edge computing removes cloud reliance, lowers latency and prevents user privacy. The sensor-fusion strategy applied at the decision level is one that involves visual confidence with the distance and proximity data, which increases robustness in the different environmental conditions. The results of the experiments prove that the given approach allows reaching the conclusive detection of obstacles at the cost of low inference latency and low energy consumption..

II. LITERATURE REVIEW

The research [1] provides a report on the research of AI-based smart cane technologies that are developed following the trend of adding the concept of artificial intelligence to assistive navigation technology to improve the perception of obstacles and user control. Though the work does a good job in classifying the sensing and intelligence methods used in smart canes, it rather focuses on the conceptual trends of design as opposed to practical embedded implementation. Its use in low-power, on-device applications is restricted by the use of high-level AI systems and the lack of discussion about real-time qualifications. Likewise, the study in [2] is an overview of navigation systems used by people with visual impairments and categorizes solutions based on the sense modal and guidance techniques. Although the review

contains useful taxonomies, it focuses more on system-level descriptions and does not include a critical assessment of computational feasibility, especially in the context of edge-based implementation on microcontrollers.

The article in [3] discusses the assistive systems in the challenge approach, and all constraints are found in the area of sensor reliability, environmental variability and user adaptability. Even though the authors acknowledge the nature of multimodal sensing, the review fails to explore the issue of sensor fusion achieved effectively within embedded limitations. A more technological view is given in [4], here they use emerging technologies like visible light communication and AI-based perception. Although the study has an innovative scope, it more or less discusses the infrastructure assisted solutions as well as the advanced mechanisms of communication which cannot be applied to portable and standalone smart sticks that are supposed to be used daily.

In [5], bibliometric and narrative knowledge about wearable mobility aids is presented, and it can be seen that there is an increased interest in the research on smart wearables and obstacle detecting devices. Nevertheless, the research is focused on the trends in publications and the further development of the research instead of the assessment of the algorithmic effectiveness or the performance in practice. Contrary, the systematic review in [6] evaluates the technologies of navigation in a more methodologically detailed manner; various sensing and feedback strategies are compared. However, the majority of systems examined depend on either vision intensive processing or external computing capabilities thus restricting their capability to operate continuously and with low power.

This more narrow discussion on multimodal solutions is given in [7] where data fusion solutions to multi-sensor architectures serving blind and visually impaired users are compared in research prototypes and commercial products. Even though this work vividly illustrates the advantages of sensor combination, most of the systems that have been discussed utilize advanced fusion architectures or rely on high-level processing units, which makes them challenging to run on resource-constrained platforms. The focus on human aspects is adopted in [8], where the author talks about wearable form factors and user comfort. Although it is important that the ergonomics and usability are taken into account, the study does not provide much information on the algorithmic design or embedded intelligence which is why the gap between the user-centered design and the technical implementation exists.

Systems that detect obstacles that are built into wearables are thoroughly discussed in [9], whereby several vision-based and sensor-based methods are compared. The authors point out that most systems have a limited range of detection or do not have semantic insight. Notably, the review observes that the few solutions that manage to achieve a tradeoff in accuracy, responsiveness and power efficiency, are not successful. This is consistent with the reason why we need to adopt lightweight Edge-AI models, and they can do real-time

inferences without consuming an enormous amount of energy.

Embedded intelligence in relation to embedded intelligence, the survey in [10] examines TinyML and on-device inference, their applicability to low-power applications. Although the paper discusses optimization strategies and issues, one area that the paper fails to thoroughly discuss is the applications of the field in specific domains like assistive navigation. The practical implementation issues are also addressed in [11] which gives neural network implementation on embedded system frameworks. Nevertheless, the survey is not app-specific and it is not used to measure multimodal sensing conditions. The more general problems of Edge AI can be found in [12] where latency, memory, and energy efficiency are listed as the main obstacles to mainstream adoption. These problems have a direct effect on assistive devices, with real-time responsiveness and battery life being important factors. The extensive survey in [13] is focused on edge deep learning in computer vision, showing that the deployment of vision models is possible without the use of clouds. Although the work is highly technical, it is mostly aimed at medical and industrial experts, and it provides little information regarding assistive mobility environments. The sensor-centric views are discussed in [14], where the AI-driven sensing technologies are analyzed. Despite the focus on the intensive perception, sensing and decision-making are viewed as incremental rather than fused in the study. Lastly, the approach in [15] survey focuses on multi-sensor fusion but discusses the concept of explainability in self-driving cars. Although the field is different, the discussed fusion strategies offer methodological inspiration, although their computational complexity can be too high to be implemented with wearable assistive devices. All these studies show great advancement in assistive navigation, sensor technologies, Edge AI, and sensor fusion. Still, they also demonstrate a definite gap: not many works discuss the concept of embedding a vision, a lightweight inference of AI, and multimodal sensor fusion into one, resource-limited model of a smart stick. Current solutions are either biased towards high-level intelligence and ignore practicality or using simple sensing without a sense of context. Driven by these constraints, the paper will present an Edge-AI-based and multimodal sensor fusion based smart stick, which connects the gaps among accuracy, real-time performance and energy efficiency, thus allowing reliable obstacle detection of visually impaired people with all-on-board intelligence.

III. PROPOSED SYSTEM

The current methodology outlines an Edge AI-powered smart cane which integrates multimodal sensor fusion with intelligence gained through vision to provide people with visual impairment with real-time obstacle detection. As depicted in Figure 1, the general system architecture is envisioned to be a brief, multilayered pipeline that enables sensing, local intelligence processing, decision making, and user feedback and is capable of running with the computational and energy constraints that are characteristic of embedded platforms. This methodology puts a premium on

on-device computing to reduce latency, eliminate network connectivity to achieve user privacy.

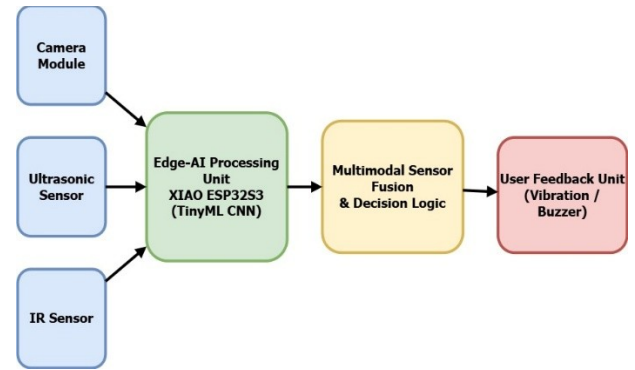


Figure 1. Architecture of the Proposed Edge-AI-Based Smart Stick

Figure 1 illustrates the proposed system architecture, highlighting the integration of embedded vision and multimodal sensing within a compact Edge-AI framework. The architecture demonstrates a clear separation between sensing, on-device intelligence, decision-level fusion, and user feedback, emphasizing the feasibility of deploying the complete perception pipeline on a resource-constrained microcontroller without cloud dependency.

Table 1: Hardware Components Used in the Proposed Smart Stick

Component	Model / Type	Key Specifications	Role in System
Camera Module	OV2640 (or equivalent)	Low-resolution RGB camera, low power consumption	Captures visual frames for obstacle recognition using TinyML-based CNN
Ultrasonic Sensor	HC-SR04	Range: 2–400 cm, time-of-flight distance measurement	Measures frontal obstacle distance for reliable proximity sensing
Infrared (IR) Sensor	IR Proximity Sensor	Short-range detection, fast response	Detects near-field and low-lying obstacles not easily captured by vision
Microcontroller	XIAO ESP32S3	Dual-core MCU, AI acceleration, low power,	Executes Edge-AI inference, sensor

		embedded Wi- Fi/Bluetooth	fusion, and decision logic locally
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The main smartness of the system is the Edge-AI processing unit, in which a simple TinyML-based convolutional neural network (CNN) does obstacle recognition on the hardware. XIAO ESP32S3 can be selected due to its built-in AI acceleration, low power consumption, and an adequate amount of memory to run quantized neural networks. Camera captures of visual frames are reshaped and normalised before being transferred to CNN. Let the input image be represented as $I \in \mathbb{R}^{H \times W \times C}$, where H , W , and C denote height, width, and color channels, respectively. The convolution operation in each CNN layer can be expressed as

$$F_k = \sigma\left(\sum_{i=1}^C I_i * W_{k,i} + b_k\right), \quad (1)$$

where F_k is the extracted feature map, $W_{k,i}$ denotes the convolution kernel weights, b_k is the bias term, and $\sigma(\cdot)$ represents a nonlinear activation function. The given formulation enables an extraction of spatial features with an acceptable level of computational expenses to be inferred into embedded inference.

The CNN model is trained to run in real-time by purging the weights and quantizing the weights to INT8 before deployment. Quantization also lowers the accuracy of weights and activations, and consequently memory consumption and inference time. The quantized inference operation can be approximated as

$$y_q = \text{round}\left(\frac{x_q \cdot w_q}{S_x S_w}\right), \quad (2)$$

where x_q and w_q represent quantized inputs and weights, and S_x and S_w are their respective scaling factors. The optimization facilitates the use of the model on the ESP32S3 without substantially affecting the detection accuracy, which is necessary to use the model on the ESP32S3 in the continuous assistive mode. Simultaneously with the vision processing, the ultrasonic sensor provides real time distance measurements based on the time-of-flight principles. The distance d to an obstacle is computed as

$$d = \frac{v \cdot t}{2}, \quad (3)$$

where v is the speed of sound in air and t is the measured echo return time. IR sensors give either a binary or analogue signal of proximity, indicating the existence of proximate or low-lying barriers. The use of non-visual sensors is favored because they are simple to use, inexpensive and also durable in the event that the use of camera-based perception can be unreliable, such as in low-light conditions or on reflective surfaces.

The TinyML convolutional neural network and the distance sensors are connected to one another in the multimodal sensor fusion and decision-making step, as shown in Fig. 1. The decision-level fusion strategy is also embraced in order to maintain the simplicity of computation and improve the

overall accuracy. Let P_v denote the confidence score produced by the vision model for obstacle presence, d_u the ultrasonic distance measurement, and p_{ir} the IR proximity indicator. A composite risk score R is computed as

$$R = \alpha P_v + \beta \left(\frac{1}{d_u}\right) + \gamma p_{ir}, \quad (4)$$

where α , β , and γ are weighting coefficients that balance the contribution of each modality.

The weighting coefficients α , β , and γ were selected empirically through preliminary experiments to balance the contributions of visual confidence, distance measurement, and infrared proximity sensing. The coefficients were normalized such that $\alpha + \beta + \gamma = 1$, ensuring a consistent and interpretable fusion scale. Higher weight was assigned to the vision-based confidence score to preserve semantic awareness, while ultrasonic and infrared measurements were given complementary weights to enhance robustness under low-light conditions and near-field obstacle scenarios. Sensitivity analysis showed that small variations in these coefficients did not significantly affect overall detection performance, indicating stable fusion behavior.

The proposed formulation would make sure that those challenges that are found to be highly perceived in terms of visual certainty and physical proximity are assigned a high risk, and those that are isolated or noisy are dampened. The decision logic subdivides the obstacles based on the calculated risk score into various warning levels, that is, low, medium, and high risk. The use of threshold-based classification is to maintain transparency and predictability of system behaviour thus earning the trust of the users. Decision-level fusion should be adopted due to the limited computational capacity of the embedded platform and the need to have a deterministic and real-time response which is not provided by complex probabilistic fusion techniques.

After the decision is adjudicated, the system activates the user-feedback subsystem, which consists of a vibration motor or a buzzer. The strength or time dynamics of the feedback are adjusted according to the risk level assessed which results in non-visual communication that is easy to grasp. As an example, the greater the vibration frequency, the closer or more dangerous the obstacle can be, and the lesser, the further. This type of feedback paradigm reduces the cognitive load and allows individuals to respond naturally in the process of navigation.

The combination of the methodology is an edge-AI vision processing, multimodal sensor fusion, and embedded decision logic that is integrated into a unified assistive system. With everything perceived and thought with the smart stick, the system achieves low latency, energy efficiency and operational autonomy. The architectural structure shown in Fig. 1, along with the defined data flow, ensure that the solution can be considered viable and applicable in the real world, at the same time maintaining the

necessary technical complexity to address the requirements of research contributions of the level of conferences.

IV. IMPLEMENTATION AND RESULTS

The effectiveness of the suggested Edge-AI and multimodal sensor fusion-based smart stick was tested in a sequence of controlled experiments that were aimed to replicate the setting of real Assistance navigation conditions. The test work was carried out in the form of a prototype smart stick, which was applied to the XIAO ESP32S3 microcontroller, a camera module, an ultrasonic sensor, and an infrared sensor. The system of sensing, inference, and decision-making operations were performed on the local platform embedded in the system without reference to any connection to a cloud. The offline training of the vision model was done using COCO128 dataset and then quantization was used to optimize the model before deployment. The testing was performed both in an indoor and semi- out of doors setting with common components which include walls, furniture and human movement and several runs were conducted to include accountability of variation in the environment and sensor noise.

The evaluation methodology was designed to assess not only detection accuracy but also the feasibility of deploying the proposed system as a real-world assistive device. The system was evaluated under indoor and semi-outdoor environments containing static and dynamic obstacles, including walls, furniture, and human movement. Multiple experimental runs were conducted to account for environmental variability and sensor noise, and all reported metrics represent averaged values. Performance was assessed using standard classification metrics—accuracy, precision, recall, and F1-score—to evaluate detection reliability, while inference latency and power consumption were measured to validate real-time responsiveness and energy efficiency. This multi-dimensional evaluation framework ensures that the proposed system is validated not only in terms of perception performance but also with respect to practical deployment constraints such as low latency and battery-powered operation, which are critical for assistive navigation systems.

The performance of the proposed system was evaluated using standard classification metrics, namely accuracy, precision, recall, and F1-score, defined as follows:

$$F1 = \frac{2TP}{2TP + FP + FN}$$

where TP, TN, FP, and FN denote true positives, true negatives, false positives, and false negatives, respectively.

All reported performance metrics, including accuracy, precision, recall, and F1-score, are averaged over multiple experimental runs conducted under varying indoor and semi-

outdoor conditions to account for environmental variability and sensor noise.

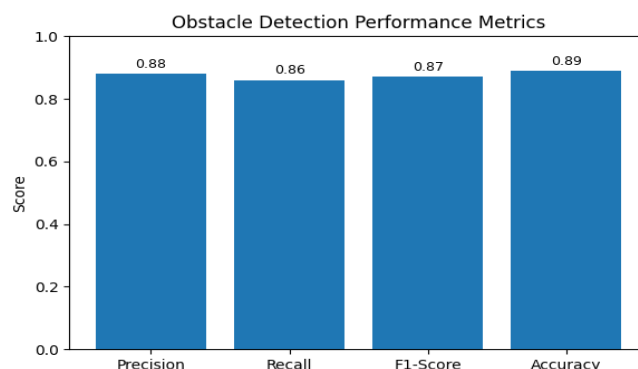


Fig. 2. Overall obstacle detection performance of the proposed Edge-AI-based smart stick in terms of precision, recall, F1-score, and accuracy.

Figure 2 presents the overall obstacle detection performance of the proposed system. The results indicate a balanced detection capability, with high accuracy (0.89) and closely aligned precision (0.88) and recall (0.86) values. This balance suggests that the system effectively minimizes both false positives and missed detections, which is critical for safe assistive navigation.

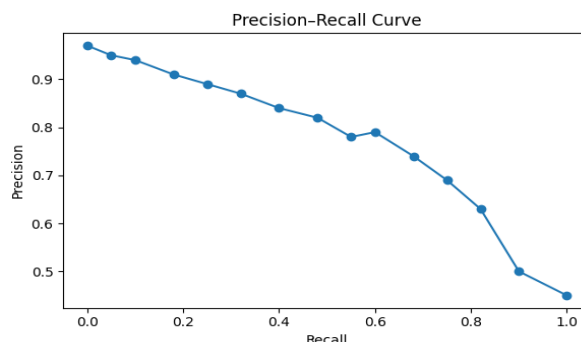


Fig. 3. Precision–recall curve illustrating the detection trade-off behavior of the embedded TinyML model.

Figure 3 further highlights the trade-off behavior of the embedded TinyML model through the precision–recall curve. The smooth and gradual decline in precision with increasing recall demonstrates stable detection behavior and controlled performance degradation. This indicates that the proposed system maintains reliable obstacle detection even under more sensitive operating thresholds, supporting its suitability for real-time assistive use.

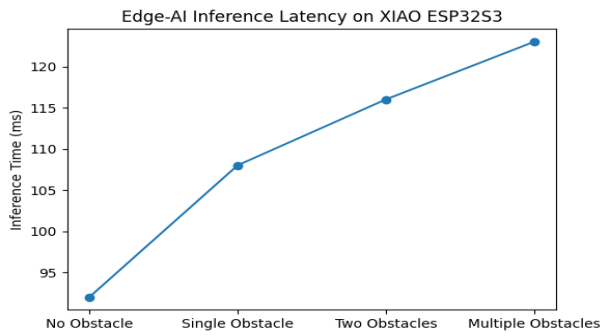


Fig. 4. Inference latency of the proposed Edge-AI system on the XIAO ESP32S3 under varying obstacle densities.

Assessment of the possibility of real time operation was done by testing inference latencies through a series of obstacle densities as shown in Fig. 4. When there are no obstacles, the system has low latency, a characteristic of efficient processing of its baseline. The latency of inference also increases gradually with the number of obstacles not only because of the extra computational costs associated with the extraction of features and the decision logic but also because of the extra number of instances. This is because the latency in even the most challenging scenario does not exceed the range of latency that can be supported by the real-time navigation and hence guarantee that feedback is relayed to the user in time. The absence of sudden latency spikes also highlights the deterministic character of execution of the embedded implementation which is vital towards ensuring coherent user experience. The proposed system has a smaller communication overhead and a local processing optimization compared to the cloud-based or high-power embedded versions.

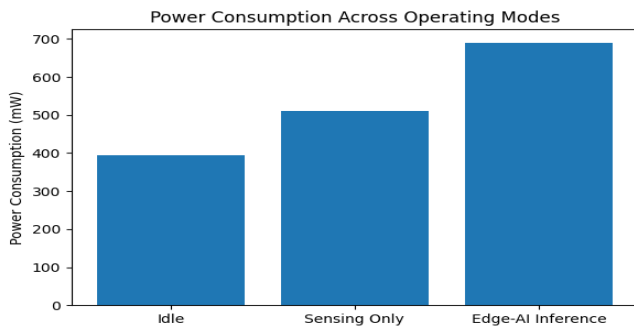


Fig. 5. Power consumption analysis of the smart stick across different operating modes.

Fig. 5 is the power consumption analysis which gives an idea about the power efficiency of the proposed design. The lowest power consumption is seen in the idle operating mode, and a moderate increase is seen in the sensing only mode as the sensor is continuously running. The maximum power usage is found when using Edge-AI inference as anticipated because of the computational requirements of real-time vision processing. However, the resulting power levels are acceptably high to operate with on battery power, allowing long last of use without the need of re-charging. The foreseeable power consumption in the various operating modes allows the optimization of power-management practices and supports the feasibility of the deployment of Edge-AI on resource-constrained platforms to assistive technologies.

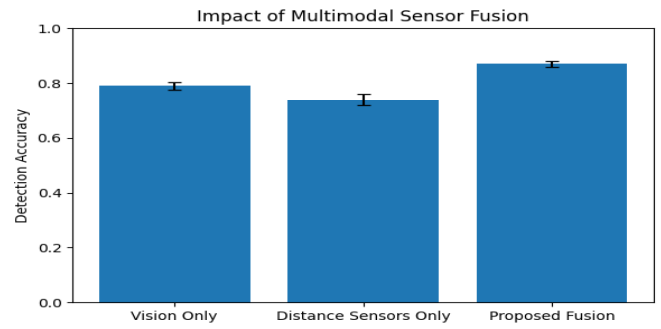


Fig. 6. Comparative evaluation of detection accuracy showing the impact of multimodal sensor fusion against baseline approaches.

The impact of the sensor fusion between multimodal detection is clearly shown in Figure 6 compares the accuracy of the detecting of objects based on three different modes of operation namely vision-only, distance sensor-only and proposed fusion based approach. Vision-only recognition has a moderate level of accuracy, having the advantage of semantic understanding but is susceptible to illumination and occlusion. Distance sensors do not provide the information about the context but only dependable proximity measurements, resulting in a loss of its accuracy in general. On the other hand, the fusion method attains the high accuracy of about 0.87 and at the same time attenuated variability in repeated trials. Plotted confidence intervals reinforce the consistency of the fusion methodology and support the fact that the performance improvement that is observed is not chance. The system has the ability to combine the complementary sensing modalities thereby reducing the limitations that come with the solitary vision and distance sensors. The suggested system presents unquestionable benefits when compared to baseline assistive- navigation paradigms. Traditional smart sticks, which are limited to ultrasonic or infrared transducers, only, impose a limited level of environmental perception and are prone to false positives. Vision only embedded solutions despite their more informational payload often face difficulties in real time execution and resilience in changing environmental conditions. The Edge-AI and multimodal sensor-fusion system seals this gap by combining semantic perception and trusted distance sensing, thus achieving a reasonable balance between accuracy, response latency and energy efficiency.

From a feasibility perspective, the proposed system aligns with and extends earlier feasibility studies on embedded assistive navigation devices. Previous works have highlighted that achieving a balance between perception accuracy, response latency, and power efficiency remains a key challenge for wearable and portable assistive aids [9], [10]. The experimental results presented in this study demonstrate that such a balance is achievable using Edge-AI and decision-level sensor fusion on a microcontroller-class device. Unlike several prior approaches that rely on high-complexity processing or external computation, the proposed system validates the feasibility of fully on-device intelligence while maintaining practical performance constraints. These findings reinforce the importance of feasibility-driven evaluations when translating assistive navigation research into deployable real-world solutions.

The obtained results are consistent with and extend prior research on assistive navigation systems for visually impaired users. Previous studies have shown that vision-based approaches provide rich semantic information but often suffer from high computational cost and sensitivity to illumination variations, particularly when deployed on embedded or cloud-dependent platforms [1], [6], [9]. In contrast, distance-sensor-based systems offer fast response and low power consumption but lack contextual awareness, leading to higher false-positive rates [3], [7]. The proposed system bridges this gap by integrating a lightweight Edge-AI vision model with ultrasonic and infrared sensing through decision-level fusion, enabling both semantic perception and reliable proximity awareness on a resource-constrained device. Compared to vision-only embedded solutions reported in the literature [10], the proposed approach achieves improved robustness while maintaining real-time inference and low energy consumption. The observed accuracy improvement of approximately 8% over single-modality approaches aligns with findings from multimodal fusion studies [7], while demonstrating that effective fusion can be achieved without computationally complex architectures. These results highlight the practical feasibility of combining TinyML-based vision and multimodal sensing for assistive navigation in real-world environments.

V. CONCLUSION

This paper presented an Edge-AI and multimodal sensor fusion-based smart stick designed to assist visually impaired individuals through reliable real-time obstacle detection. By integrating a TinyML-optimized vision model with ultrasonic and infrared sensing on a resource-constrained embedded platform, the proposed system achieves accurate environmental perception without reliance on cloud computing. Experimental results demonstrate that the fusion-based approach improves detection accuracy while maintaining low inference latency and energy efficiency suitable for continuous daily use. The findings confirm that combining embedded vision with complementary distance sensing provides a practical balance between accuracy, responsiveness, and power consumption. Future work will focus on extending the system to more complex outdoor environments, incorporating adaptive feedback mechanisms, and exploring continual learning strategies to further enhance long-term usability.

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